

Jeffrey Hu

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EDUCATION

University of Cambridge

PhD student

2025–present
Cambridge, UK

Supervisor: Professor Ayush Tewari.

Massachusetts Institute of Technology

Master of Science, 2017–2019

2014–2019
Cambridge, MA

Bachelor of Science, 2014–2018.

Supervisor: Professor Antonio Torralba.

RESEARCH & OPEN SOURCE

Inria — GraphDeco

Research Engineer

Feb 2025–Sep 2025
Nice, France

- Worked with Professor George Drettakis on differentiable rendering with 3D Gaussian splats.
- Developed methods for intrinsic decomposition and editable physically-based reflections.

gsplat

Open-source Contributor

From Apr 2024

- Contributed to an open-source Gaussian splatting library for 3D reconstruction and novel-view synthesis, including:
 - 3DGS-MCMC for improved densification and bilateral grids for handling exposure changes.
 - Combined Self-Organizing Gaussian Splats and spherical-harmonics k-means clustering for Gaussian splat compression.
 - Fisheye camera support, camera defocus blur modeling, and 2.5D Gaussian splatting for surface reconstruction.

PUBLICATIONS

Intrinsic Decomposition and Editing of 3D Gaussian Splats

A. Lanvin, **J. Hu**, S. Lucas, A. Bousseau, and G. Drettakis

I3D / Proceedings of the ACM on Computer Graphics and Interactive Techniques, 2026.

Adaptive Spatio-Temporal 3D Gaussian Splatting for Scenes with Oscillatory Motion

P. Tzathas, **J. Hu**, A. Meuleman, G. Cordonnier, and G. Drettakis

Eurographics / Computer Graphics Forum, 2026.

Efficient Camera-Controlled Video Generation of Static Scenes via Sparse Diffusion and 3D Rendering

J. Chen, **J. Hu**, J. Lasenby, and A. Tewari

arXiv preprint, 2026.

Editable Physically-based Reflections in Raytraced Gaussian Radiance Fields

Y. Poirier-Ginter, **J. Hu**, J.-F. Lalonde, and G. Drettakis

SIGGRAPH Asia, 2025.

gsplat: An Open-Source Library for Gaussian Splatting

V. Ye, R. Li, J. Kerr, M. Turkulainen, B. Yi, Z. Pan, O. Seiskari, J. Ye, **J. Hu**, M. Tancik, and A. Kanazawa

Journal of Machine Learning Research, 2025.

WORK EXPERIENCE

Parallel Systems*Senior Software Engineer*

Mar 2022–Apr 2024

Los Angeles, CA

- Led development of a real-time multitask computer vision network and data engine for an autonomous train perception system.
- Iteratively built a multitask network for 3D object detection, 3D track prediction, depth prediction, and bird's-eye-view segmentation.
 - Wrote a PyTorch library for training models with various backbones, task-specific heads, and loss functions.
 - Developed a representation for railroad track geometry by regressing 3D splines and classifying switch states.
 - Shipped multiple model iterations onto a Jetson using Rust, TensorRT, and DeepStream.
- Developed a 3D reconstruction and autolabeling pipeline for scalable ground-truth generation using Dagster and Kubernetes.
 - Used zero-shot, open-vocabulary foundation models for object detection, segmentation, tracking, and monocular depth estimation.
 - Aggregated predictions across time to reconstruct 3D scene representations as high-quality labels for training and evaluating the real-time network.
 - Built infrastructure to balance datasets, mine scenarios, and automatically generate test reports to support the regulatory safety case over time.
- Contributed to sensor calibration, onboard data collection, triggering, and upload infrastructure.

TuSimple*Research Engineer II*

Jan 2020–Mar 2022

San Diego, CA

Amazon Prime Air*Software Engineer Intern*

Jun 2016–Aug 2016

Seattle, WA

LEADERSHIP & INTERESTS

MIT Varsity Squash — Captain

Sep 2014–Jun 2018

Programming: Python, PyTorch, C++, CUDA.**Interests:** Squash, cycling, skiing, hiking.